

The derivative of a composite function

Lessons 10–12 · Algebra 11, §1.5 · P1 · 7.5 · 28 problems

Differentiate with respect to x unless a rate is asked for. Factorise every answer — the factorised form is the one later questions need.

Easy 7

One bracket, one chain. The whole test is whether the second factor — the derivative of the inside — is there at all.

$$1 \quad ((x+5)^3)' \quad \left| \quad 3(x+5)^2 \right|$$

The inside is $x+5$, whose derivative is 1, so the second factor changes nothing.

$$2 \quad ((4x-1)^2)' \quad \left| \quad 8(4x-1) \right|$$

$2(4x-1) \times 4 = 8(4x-1)$.

$$3 \quad ((1-3x)^5)' \quad \left| \quad -15(1-3x)^4 \right|$$

The inside derivative is -3 , so $5(1-3x)^4 \times (-3)$. The sign is the whole point.

$$4 \quad (\sqrt{5x})' \quad \left| \quad \frac{5}{2\sqrt{5x}} \right|$$

Write it as $(5x)^{1/2}$. Then $\frac{1}{2}(5x)^{-1/2} \times 5 = \frac{5}{2\sqrt{5x}}$.

$$5 \quad ((x^3+2)^4)' \quad \left| \quad 12x^2(x^3+2)^3 \right|$$

$4(x^3+2)^3 \times 3x^2$.

$$6 \quad \left(\frac{1}{(3x+1)^2} \right)' \quad \left| \quad -\frac{6}{(3x+1)^3} \right|$$

Write it as $(3x+1)^{-2}$. Then $-2(3x+1)^{-3} \times 3 = -6(3x+1)^{-3}$.

$$7 \quad ((2-x^2)^6)' \quad \left| \quad -12x(2-x^2)^5 \right|$$

$6(2-x^2)^5 \times (-2x)$.

Medium 7

The inside is no longer linear, so the second factor carries real work. In 12 the outside is a power but the inside needs the product rule first; 14 is a rate, so differentiate before any number goes in.

$$8 \quad ((2x^2+3x)^5)' \quad \left| \quad 5(4x+3)(2x^2+3x)^4 \right|$$

Inside derivative $4x+3$, so $5(2x^2+3x)^4(4x+3)$.

$$9 \quad (\sqrt[3]{x^2+1})' \quad \left| \quad \frac{2x}{3(x^2+1)^{2/3}} \right|$$

Write it as $(x^2+1)^{1/3}$. Then $\frac{1}{3}(x^2+1)^{-2/3} \times 2x$.

10 $\left(\frac{1}{\sqrt{4-x}}\right)' \quad \left| \quad \frac{1}{2(4-x)^{3/2}} \right|$

$(4-x)^{-1/2}$ differentiates to $-\frac{1}{2}(4-x)^{-3/2} \times (-1)$. Two minus signs, so the answer is positive.

11 $\left(x + \frac{1}{x}\right)^3' \quad \left| \quad \frac{3(1 - \frac{1}{x^2})(x + \frac{1}{x})^2}{x^2} \right|$

The inside is $x + x^{-1}$, whose derivative is $1 - x^{-2}$.

12 $(x^2(3x-1)^4)' \quad \left| \quad \frac{2x(3x-1)^3(9x-1)}{x^2} \right|$

Product rule, with the chain rule on the second factor: $2x(3x-1)^4 + 12x^2(3x-1)^3$. Take out $2x(3x-1)^3$, leaving $(3x-1) + 6x = 9x-1$.

13 $(\sqrt{x^2-4x})' \quad \left| \quad \frac{x-2}{\sqrt{x^2-4x}} \right|$

$\frac{1}{2\sqrt{x^2-4x}} \times (2x-4)$, and the 2 cancels.

14 The radius of a circle shrinks at 0.5 cm/s. Find $\frac{dA}{dt}$ when $r = 6$ cm. $\left| \quad -6\pi \text{ cm}^2/\text{s} \right|$

$A = \pi r^2$, so $\frac{dA}{dr} = 2\pi r = 12\pi$. Then $\frac{dA}{dt} = 12\pi \times (-0.5) = -6\pi$. The sign is negative because the circle is shrinking.

Hard 7

Two rules at once. Decide the outermost structure first — product, quotient or power — and only then reach for the chain rule inside it.

15 Differentiate $y = (x-1)^3(x+2)^2$ and factorise the answer. $\left| \quad (x-1)^2(x+2)(5x+4) \right|$

Product rule: $3(x-1)^2(x+2)^2 + 2(x-1)^3(x+2)$. The common factor is $(x-1)^2(x+2)$, and what is left is $3(x+2) + 2(x-1) = 5x+4$.

16 Differentiate $y = \frac{(2x+3)^4}{x^2}$. $\left| \quad \frac{2(2x+3)^3(2x-3)}{x^3} \right|$

Quotient rule: the numerator is $8(2x+3)^3x^2 - 2x(2x+3)^4$. Take out $2x(2x+3)^3$, leaving $4x - (2x+3) = 2x-3$. One x cancels with x^4 .

17 Differentiate $y = \sqrt{(1+\sqrt{x})}$. $\left| \quad \frac{1}{4\sqrt{x}\sqrt{(1+\sqrt{x})}} \right|$

Chain rule twice: $\frac{1}{2\sqrt{(1+\sqrt{x})}} \times \frac{1}{2\sqrt{x}}$. The outer root is differentiated first, with the inside left alone.

18 Find the gradient of $y = (x^2+1)^3$ at $x = 1$. $\left| \quad 24 \right|$

$y' = 6x(x^2+1)^2$. At $x = 1$ this is $6 \times 1 \times 4 = 24$.

19 Find the x -coordinates of the stationary points of $y = (x^2-9)^4$. $\left| \quad x = 0 \text{ and } x = \pm 3 \right|$

$y' = 8x(x^2-9)^3$. A product is zero when a factor is zero, so $x = 0$ or $x^2 = 9$. Three stationary points, not one.

20 Differentiate $y = \frac{x}{\sqrt{x^2 + 1}}$. $\left| \frac{1}{(x^2 + 1)^{3/2}} \right|$

Quotient rule: $[\sqrt{x^2 + 1} - x \cdot \frac{x}{\sqrt{x^2 + 1}}]/[x^2 + 1]$. Multiply top and bottom by $\sqrt{x^2 + 1}$: the numerator becomes $(x^2 + 1) - x^2 = 1$.

21 The surface area of a cube grows at 12 cm²/s. How fast is the side growing when the side is 5 cm?

0.2 cm/s

$S = 6a^2$, so $\frac{dS}{da} = 12a = 60$. From $\frac{dS}{dt} = \frac{dS}{da} \times \frac{da}{dt}$, $12 = 60 \times \frac{da}{dt}$, so $\frac{da}{dt} = \frac{1}{5}$.

Very hard 7

Letters instead of numbers, and questions that ask why. In 26 the discriminant decides how many answers there are; in 28 the inside never reaches zero, which is what makes the stationary point unique.

22 Show that $(ax + b)^n$ differentiates to $an(ax + b)^{n-1}$ for every constant a , b and every index n . shown

Let $u = ax + b$, so $y = u^n$. Then $\frac{dy}{du} = nu^{n-1}$ and $\frac{du}{dx} = a$. Multiplying, $\frac{dy}{dx} = nu^{n-1} \times a = an(ax + b)^{n-1}$. This is the shortcut used silently in every easy question above.

23 Differentiate $y = \left(\frac{x+1}{x-1}\right)^3$. $\left| -\frac{6(x+1)^2}{(x-1)^4} \right|$

The outermost structure is a power, so chain first. With $u = \frac{x+1}{x-1}$, the quotient rule gives

$\frac{du}{dx} = \frac{(x-1) - (x+1)}{(x-1)^2} = -\frac{2}{(x-1)^2}$. Then $3u^2 \times \frac{du}{dx} = \frac{3(x+1)^2}{(x-1)^2} \times \left(-\frac{2}{(x-1)^2}\right)$.

24 Find the equation of the tangent to $y = \sqrt{3x+4}$ at $x = 4$. $\left| y = \frac{3}{8}x + \frac{5}{2} \right|$

$y(4) = \sqrt{16} = 4$, so the point is $(4, 4)$. $y' = \frac{3}{2\sqrt{3x+4}}$, which at $x = 4$ is $\frac{3}{8}$. Then $y - 4 = \frac{3}{8}(x - 4)$. Check: at $x = 4$, $\frac{3}{8}(4) + \frac{5}{2} = 4$. ✓

25 A spherical balloon is inflated at 100 cm³/s. How fast is its radius growing when $r = 5$ cm?

$\left| \frac{1}{\pi} \approx 0.318 \text{ cm/s} \right|$

$V = \frac{4}{3}\pi r^3$, so $\frac{dV}{dr} = 4\pi r^2 = 100\pi$. From $\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$, $100 = 100\pi \times \frac{dr}{dt}$, giving $\frac{dr}{dt} = \frac{1}{\pi}$. Note that the answer is a rate of length, so the units are cm/s, not cm³/s.

26 $f(x) = (x^2 + k)^3$ has gradient 36 at $x = 1$. Find the possible values of k . $k = -1 \pm \sqrt{6}$

$f'(x) = 6x(x^2 + k)^2$, so $f'(1) = 6(1 + k)^2 = 36$, giving $(1 + k)^2 = 6$. A square equals a positive number in two ways, so $1 + k = \pm\sqrt{6}$ and there are **two** values of k .

- 27 Sand falls into a conical pile whose height always equals the radius of its base, at $8 \text{ cm}^3/\text{s}$. How fast is the height rising when $h = 4 \text{ cm}$? $\frac{1}{2\pi} \approx 0.159 \text{ cm/s}$

Use the condition to remove r before differentiating: with $r = h$, $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi h^3$. Then $\frac{dV}{dh} = \pi h^2 = 16\pi$ and $8 = 16\pi \times \frac{dh}{dt}$, so $\frac{dh}{dt} = \frac{1}{2\pi}$. Differentiating before the substitution leaves two unknown rates and gets nowhere.

- 28 Show that $y = (x^2 - 2x + 2)^3$ has exactly one stationary point, and find it. $(1, 1)$

$y' = 3(x^2 - 2x + 2)^2(2x - 2) = 6(x - 1)(x^2 - 2x + 2)^2$. Completing the square, $x^2 - 2x + 2 = (x - 1)^2 + 1 \geq 1$, so that factor is never zero. Hence $y' = 0$ only when $x = 1$, and $y(1) = 1^3 = 1$.